

A Parameter-Curve Search for Solutions of

$$a^4 + b^4 + c^4 = d^4$$

GPT 5.5

May 2026

Abstract

This note describes a computational search for integer solutions of

$$a^4 + b^4 + c^4 = d^4.$$

The calculation used explicit rational-parameter families associated with the Elkies–Tomita–Piezas style construction, rather than a blind search in the four original variables. A 64-shard OpenMP run over five such families found 13 distinct primitive equations with $d \leq 2 \cdot 10^{10}$, including the largest five displayed below. Every reported equation was verified with exact integer arithmetic.

1 Background

The equation

$$a^4 + b^4 + c^4 = d^4$$

is the classical counterexample setting for Euler’s fourth-power conjecture. Frye’s Connection Machine computation found the small solution

$$95800^4 + 217519^4 + 414560^4 = 422481^4$$

and is still useful as a model for equation-specific pruning and parallel search design [9]. Elkies gave an elliptic-curve construction which produces infinitely many coprime natural-number solutions and includes, among others,

$$2682440^4 + 15365639^4 + 18796760^4 = 20615673^4$$

[7]. The broader computational literature on equal sums of like powers and sorted/meet-in-the-middle enumeration provides the surrounding algorithmic context [2, 6, 10, 11].

The present computation follows the parametric rather than exhaustive route: instead of searching a four-dimensional box in a, b, c, d , it searches rational parameters on several genus-one-derived families and then verifies the resulting integer identities exactly.

2 Algorithm

For a rational parameter

$$v = \frac{p}{q}, \quad \gcd(p, q) = 1, \quad q > 0,$$

each curve supplies three quadratic forms

$$\begin{aligned} A(p, q) &= a_0q^2 + a_1pq + a_2p^2, \\ B(p, q) &= b_0q^2 + b_1pq + b_2p^2, \\ D(p, q) &= d_0q^2 + d_1pq + d_2p^2, \end{aligned}$$

and one quartic form

$$F(p, q) = f_0q^4 + f_1pq^3 + f_2p^2q^2 + f_3p^3q + f_4p^4.$$

The curve also has a small integer multiplier k . A parameter is retained only when

$$kF(p, q)$$

is a perfect square. In that case define

$$C(p, q) = \sqrt{kF(p, q)}.$$

The candidate integer quadruple is

$$(|A(p, q)|, |B(p, q)|, |C(p, q)|, |D(p, q)|).$$

It is divided by the common gcd of all four entries, and the resulting primitive quadruple is checked by exact multiprecision arithmetic:

$$a^4 + b^4 + c^4 = d^4.$$

No floating point arithmetic is used in the final verification.

2.1 Modular filtering

Perfect-square testing on every (p, q) pair would be unnecessarily expensive. The implementation first checks whether $kF(p, q)$ is a quadratic residue modulo a stack of primes:

1009, 1013, 1019, 1021, 1031, 1033, 1039, 1049, 1051, 1061, 1063, 1069, 1087, 1091, 1093, 1097.

Only parameters passing all residue tests are subjected to the exact integer square-root test. This preserves correctness: the modular tests can reject non-squares, but cannot create false exact hits.

3 Curves Searched

The five explicit families used in the run are listed below. They are presented in homogeneous p, q form so that integer arithmetic can be used throughout.

Curve C0.

$$\begin{aligned} k &= 4, \\ A &= -313q^2 + 484pq + 85p^2, \\ B &= 10q^2 - 586pq + 68p^2, \\ D &= -363q^2 + 204pq - 357p^2, \\ F &= 22030q^4 + 28849pq^3 - 56158p^2q^2 + 36941p^3q - 31790p^4. \end{aligned}$$

Curve C1.

$$\begin{aligned}
k &= 4, \\
A &= 15968q^2 - 2334pq - 59p^2, \\
B &= 7068q^2 + 3082pq + 10p^2, \\
D &= 22628q^2 + 54pq + 159p^2, \\
F &= 110301312q^4 + 10244932pq^3 - 1285119p^2q^2 + 5299p^3q - 6260p^4.
\end{aligned}$$

Curve C2.

$$\begin{aligned}
k &= 1, \\
A &= 11980q^2 - 1673pq - 54p^2, \\
B &= -36q^2 + 2321pq - 3p^2, \\
D &= 24677q^2 + 203pq + 71p^2, \\
F &= 591800025q^4 + 20030510pq^3 + 1671327p^2q^2 + 92762p^3q - 4112p^4.
\end{aligned}$$

Curve C3.

$$\begin{aligned}
k &= 1, \\
A &= -1058960q^2 + 11203324pq + 178500100p^2, \\
B &= 518320q^2 + 16483396pq - 294176372p^2, \\
D &= 1304433q^2 - 27003006pq + 345712797p^2, \\
F &= -1251219988511q^4 + 78204922436804pq^3 \\
&\quad - 1649103906705762p^2q^2 + 19988050672538996p^3q \\
&\quad - 76026722992074935p^4.
\end{aligned}$$

Curve C4.

$$\begin{aligned}
k &= 1, \\
A &= -20150032q^2 + 34614497pq + 67829197p^2, \\
B &= 5444488q^2 + 58502527pq - 148896448p^2, \\
D &= 30141789q^2 - 124521519pq + 197120781p^2, \\
F &= -812211871484873q^4 + 7722674874928166pq^3 \\
&\quad - 27864960882719827p^2q^2 + 48860237516933014p^3q \\
&\quad - 31577722089576368p^4.
\end{aligned}$$

4 Computation

The production run searched the following range:

$$|p| \leq 15,000,000, \quad 1 \leq q \leq 1,500,000, \quad d \leq 20,000,000,000.$$

The denominator interval was split into 64 independent shards. Each shard ran on one 96-core Cascadelake node using OpenMP parallelism. All 64 shards completed, and all five curves were completed on every shard.

Quantity	Observed value
Curve-shards completed	320/320
Shard elapsed time	2:57:46 to 3:08:08
Memory use	about 15–28 MB per task
Parameter hit lines	18
Distinct primitive equations	13

Different parameter values can map to the same primitive equation. The distinct-equation count therefore deduplicates by the sorted triple (a, b, c) and by d , not by the parameter p/q .

5 Five Largest Equations Found

Interpreting “largest” by the value of d , the five largest distinct equations found in the run were:

Rank	Curve and parameter	a	b	c, d
1	C1, 13617482/1280007	140976551	5821981400	15355831360, 15434547801
2	C0, 184653/394349 or 30671/229738	664793200	2448718655	3134081336, 3393603777
3	C3, 10691/353335	686398000	1237796960	1662997663, 1787882337
4	C1, $-50191/8685$	50237800	632671960	1670617271, 1679142729
5	C0, $-3015/9707$	219076465	275156240	630662624, 638523249

Written explicitly, these are:

$$\begin{aligned}
140976551^4 + 5821981400^4 + 15355831360^4 &= 15434547801^4, \\
664793200^4 + 2448718655^4 + 3134081336^4 &= 3393603777^4, \\
686398000^4 + 1237796960^4 + 1662997663^4 &= 1787882337^4, \\
50237800^4 + 632671960^4 + 1670617271^4 &= 1679142729^4, \\
219076465^4 + 275156240^4 + 630662624^4 &= 638523249^4.
\end{aligned}$$

6 Interpretation

The calculation does not prove absence of solutions outside these families or outside the parameter bounds. Its point is algorithmic: known algebraic structure turns a very sparse four-variable Diophantine search into many independent rational-parameter scans with exact verification at the end. This complements the exhaustive and meet-in-the-middle traditions represented by Frye’s Connection Machine search and Bernstein’s sorted-sums method [2, 9].

The diagonal quartic surface literature gives the broader geometric context: these equations are rational-point problems on diagonal quartic surfaces, and there is a substantial body of work on constructing, searching, and explaining rational points on such surfaces [3–5, 8, 12]. Recent work on related Euler-surface parametrization problems gives another source of explicit rational-map workflows [1].

References

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